

## Project 1: Student Performance Analyzer

### Algorithm

1. Start
2. Initialize student database.
3. Display menu.
4. Read user's choice.
5. If choice = Add Student:
  - Input roll number and name.
  - Check if roll number already exists.
  - If yes, display error.
  - Else, store student details.
6. If choice = Enter Marks:
  - Input roll number.
  - Verify student exists.
  - Input marks for each subject.
  - Validate marks (0–100).
  - Store marks.
7. If choice = Student Report:
  - Input roll number.
  - Calculate average.
  - Determine grade.
  - Determine pass/fail status.
  - Display report.
8. If choice = Subject Toppers:
  - Find highest mark in each subject.
  - Display topper names.
9. If choice = Overall Ranking:
  - Calculate average of all students.
  - Apply Bubble Sort.
  - Display ranks.
10. If choice = Summary Report:

- Count total students.
- Count pass and fail students.
- Calculate pass percentage.
- Display summary.

11. If choice = Exit:

- Stop program.

12. End.

### Flowchart

